

AY2627

HHS4185 Common Rehabilitation Conditions

HHS4185J 復康科常見疾病

Workshop 工作坊

WS2

Introduction to Rehabilitation Equipment (2)

復康器材簡介 (2)

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Content 內容

- Able to recognize & operate common equipment in rehabilitation settings

- ❖ Ceiling Hoist / Mobile Hoist

- ❖ Tilt Table / Easy Stand

- ❖ Aids for activities of daily living (ADL)

- 能夠識別和操作復康環境中的常用設備

- ❖ 天花吊機/ 活動式吊機

- ❖ 傾斜床/ 易站儀

- ❖ 日常生活活動輔助工具 (ADL)

Ceiling Hoist / Portable Hoist

天花吊機 / 活動式吊機

- A hoist is a mechanical lift or hydraulic lift, that can be free standing or attached to a ceiling
- Used to move patients
 - ✓ who cannot **bear weight**
 - ✓ who are **unpredictable** or **unreliable**
 - ✓ who have a medical condition that does **not allow them to stand or assist with moving**
 - ✓ e.g., **Spinal Cord Injury (lower limbs paralyzed, tetraplegia)**

- 吊升機是一種機械升降或液壓升降裝置，可以是獨立式或固定在天花板上。
- 用於移動：
 - ✓ **無法承重**的病人
 - ✓ **不可預測**或**不可靠**的病人
 - ✓ 有醫療狀況**無法站立或協助移動**的病人
 - ✓ 例子：脊椎受損(下肢癱瘓、四肢癱瘓)



Ceiling Hoist / Mobile Hoist

天花吊機 / 活動式吊機(移位機)

Important notes

- Assessed and assigned by physiotherapists
- No hip and pelvic fracture or problems
- Must operated by 2 people(1 x main operation of the hoist, 1 x supervision of patient)

重點

- 由物理治療師評估並指派
- 沒有髖關節和骨盆骨折或問題
- 必須由兩人操作（**1**人負責吊機操作，**1**人監察病人）

Key Steps 重點步驟

A

- ❖ Environment and hoist preparation

B

- ❖ Patient postural preparation

C

- ❖ Move the patient
 - Moved from the original location to another place, usually
 - Bed to wheelchair or
 - Wheelchair to bed

D

Resume patient in comfortable & supported position

A

- ❖ 環境及吊機的準備

B

- ❖ 病人的姿位準備

C

- ❖ 轉動移病人
 - 由原本的地方轉移至另一地方通常
 - 床到輪椅或
 - 輪椅到床

D

- ❖ 讓病人恢復舒適且有支撐的位置

A Environment and hoist preparation

環境及吊機的準備

Preparation in advance

Check if the transfer machine is operating normally?

Are the batteries installed and charged?

Is the controller operating normally?

Is the lifting bracket stable?

Is the floor stand opening and closing normal?

Is the implementation smooth?

Check if the sling is intact?

Is there enough space on site to operate the transfer machine?

Is the patient suitable for being moved, including their physical condition and emotional expression?

事前預備

❖ 檢查移位機是否操作正常？

- 電池是否已安裝好及充電？
- 控制器是否操作正常？
- 起重支架是否穩固？
- 坐地支架是否開合正常？
- 推動是否暢順？

❖ 檢查吊兜是否完好無缺？

❖ 現場是否有足夠空間操作移位機？

❖ 病人是否適合被移動位置，包括身體狀況及情緒表現？

B. Patient postural preparation (Wheelchair to bed)

病人的姿位準備(輪椅至床)

Starting from the seats

1. Place the wheelchair/training chair and bed in the appropriate position so that there is enough space to operate the repositioner, and also lock the wheelchair/bed wheels.
2. Release the patient's safety belt/restraints, assist the patient in leaning forward, pull the sling bag down as much as possible to wrap the patient's head and body propulsion, then help the patient lean back in the chair.
3. Wear the sling on both sides of the thighs, and wrap the patient's thighs from the outside in, bottom to top. It is necessary to pull the sling strap upward as much as possible to reduce the space at the bottom of the sling sling for the patient's hip.

由座位開始

1. 將輪椅／訓練椅及床擺放在合適位置，令使用移位機時有足夠空間操作，同時要把輪椅／床的轆鎖上。
2. 解開病人的安全帶／約束物品，協助病人身體前傾，把吊兜盡量向下拉將包好病人頭及身軀部份，再協助病人後倚回椅背。
3. 把吊兜帶在大腿兩側，由外至內、下至上包好病人的大腿。需要盡量把吊兜帶向上拉，以便縮細病人臀部在吊兜底部的空間。

B. Patient postural preparation (Wheelchair to bed)

病人的姿位準備(輪椅至床)

4. Push the transfer machine in front of the patient and open the floor support.
 5. When lowering the lifting bracket to an appropriate height, pay attention to whether the patient's head touches the lifting bracket.
 6. Hook the four hanging loops of the hanging sling hoop, making sure the upper and lower halves are hooked in the same color.
 7. Assist the patient with both hands in the chest/sling pocket space and inform the patient that their body will rise and transfer to the bed, training chair, or wheelchair.
4. 把移位機推至病人前面，並打開坐地支架。
 5. 下降起重支架到適當高度，需留意病人頭部會否碰到起重支架。
 6. 鈎好吊兜四個吊圈，留意上、下半部的吊圈必須鈎著相同顏色。
 7. 協助病人雙手放在胸前／吊兜空間內，並告知病人身體將會升起及轉移到床／訓練椅／輪椅上。

C. Move the patient(Wheelchair to bed)

轉動移病人(輪椅至床)

8. Once all the hanging rings are securely fastened, raise the patient to the appropriate height according to the system.
 9. Shrink the floor support frame as needed, and the pushing machine can reach above the bed, training chair, or wheelchair.
 10. During the procedure, attention must be paid to the patient's emotional changes and whether any parts of the body touch other objects.
 11. Slowly lower the patient onto the chair/bed.
8. 再次確定所有吊圈扣穩後，按制升起病人至合適高度。
 9. 按需要縮細坐地支架，推移位機到達床／訓練椅／輪椅上方。
 10. 在整過移動過程中必須留意病人情緒變化及身體各部份有沒有碰到其他物件。
 11. 緩緩下降病人到椅／床上。

D Resume patient in comfortable & supported position(W/C to bed)

恢復病人舒適且有支撐的體位(輪椅至床)

12. Remove all the hanging rings and push the transfer machine open.
13. Then remove the sling bag wrapping the patient, assist the patient in sitting/lying down, and fasten the safety belt or restraints.
14. Put all equipment back in their proper places.
15. On the bed
16. Assist the patient in turning to one side and folding the sling bag to push toward the patient.
17. Assist the patient in turning to the other side and pulling out the sling.
18. Assist the patient in lying down and putting on restraints.

12. 除下所有吊圈，推開移位機。

13. 再除下包著病人的吊兜，協助病人坐／躺好及扣上安全帶／約束物品。

14. 把所有器材放回適當位置。

在床上

15. 協助病人轉身向一面，把吊兜摺起推向病人。

16. 協助病人轉身至另一面，把吊兜拉出。

17. 協助病人躺好及穿上約束物品。

B. Patient postural preparation(Bed to W/C)

病人的姿位準備(床至輪椅)

1. Place the bed and wheelchair/training chair in appropriate positions so that there is enough space to operate the transfer machine, and also lock the wheelchair/bed wheels.
2. Raise the bed to the appropriate height and unfasten the patient's restraints.
3. Assist the patient in turning to one side, then fold half of the sling from the other side to the patient's body. Then help the patient turn to the other side and pull out the other half of the sling to return the patient to a flat position.
4. Wear the sling on both sides of the thighs, and wrap the patient's thighs from the outside in, bottom to top. It is necessary to pull the sling strap upward as much as possible to reduce the space at the bottom of the sling sling for the patient's hip.

1. 將床及輪椅／訓練椅擺放在合適位置，令使用移位機時有足夠空間操作，同時要把輪椅／床的轆鎖上。
2. 把床升高到合適高度，解開病人的約束物品。
3. 協助病人轉身向一面，從另一面鋪下摺起一半的吊兜到病人身體下。再協助病人轉身到另外面，並拉出另外一半吊兜，將病人回復至平臥位置。
4. 把吊兜帶在大腿兩側，由外至內、下至上包好病人的大腿。需要盡量把吊兜帶向上拉，以便縮細病人臀部在吊兜底部的空間。

B. Patient postural preparation(Bed to W/C)

病人的姿位準備(床至輪椅)

5. Push the transfer machine from the side into the underbed, that is, the lifting bracket is placed above the patient's body.
 6. When lowering the lifting bracket to an appropriate height, pay attention to whether any part of the patient's body touches the lifting bracket.
 7. Fasten the four hanging hoops of the hanging shoe securely, making sure the upper and lower halves are the same color.
 8. Assist the patient in placing both hands in the chest/sling pocket space and inform them that their body will rise and transfer to the training chair/wheelchair.
5. 把移位機從側面推入床下底，即起重支架在病人身體上方。
 6. 下降起重支架到適當高度，需留意病人身體部份會否碰到起重支架。
 7. 扣好吊兜四個吊圈，留意上、下半的吊圈必須是相同顏色。
 8. 協助病人雙手放在胸前／吊兜空間內，並告知病人身體將會升起及轉移到訓練椅／輪椅上。

C. Move the patient(Bed to W/C)

轉動移病人(床至輪椅)

9. Once all the hanging rings are securely fastened, raise the patient to the appropriate height according to the system.
 10. When the transfer machine reaches the training chair/wheelchair, the seat support is opened, then pushed over the chair/wheelchair to slowly lower the patient.
 11. During movement, attention must be paid to the patient's emotional changes and whether any parts of the body are injured.
 12. Remove all the hanging rings and push the transfer machine open.
9. 再次確定所有吊圈扣穩後，按制升起病人至合適高度。
 10. 推移位機到達訓練椅／輪椅前，打開坐地支架，再推至訓練椅／輪椅上方，緩緩下降病人。
 11. 移動過程中必須留意病人情緒變化及身體各部份有沒有碰傷。
 12. 除下所有吊圈，推開移位機。

D Resume patient in comfortable & supported position (Bed to W/C)

恢復病人舒適且有支撐的體位(床至椅)

13. Then remove the sling bag wrapping the patient, assist the patient in sitting properly, and fasten the safety belt or restraints.
14. Put all equipment back in their proper places.

Remove the sling

In the seat

15. Assist the patient in lifting their thighs and remove the sling covering the thigh from top to bottom and inside out.
16. Assist the patient in leaning forward and pull out the sling from behind.
17. Assist patients in sitting properly and fastening seatbelts/restraints.

13. 再除下包著病人的吊兜，協助病人坐好及扣上安全帶／約束物品。

14. 把所有器材放回適當位置。

除吊兜

在座位上

15. 協助病人大腿抬起，從上至下、內至外除下包著大腿的吊兜。
16. 協助病人前傾，從後向上拉出吊兜。
17. 協助病人坐好及扣上安全帶／約束物品。



電動輔助起置器
的使用要點

Notes in using
electrical hoist

<https://www.youtube.com/watch?v=08OgyL9VwTM>

Tilt Table 傾斜床(企床)/ Easy Stand 易站儀

To minimise the adverse effects of prolonged immobilization

- orthostatic hypotension
- reduced oxygen consumption
- venous pooling
- reduced lung volumes
- impaired gas exchange
- muscle atrophy
- joint contractures
- peripheral nerve injuries, and pressure areas
- to reintroduce patients to the vertical position.

最小化長期不活動的負面影響

- 直立性低血壓 / 姿勢性低血壓
- 減少氧氣消耗
- 靜脈淤積
- 肺容積減少
- 氣體交換受損
- 肌肉萎縮
- 關節攣縮
- 周邊神經損傷和壓力區域
- 讓病人重新適應立直姿勢



Tilt Table 傾斜床/ Easy Stand 易站儀

Benefits include

- increased ventilation
- increased arousal
- improved weight bearing of the lower limbs
- facilitation of antigravity exercise of the limbs
- promote vasomotor training

好處包括

- 增加換氣
- 增加覺醒
- 改善下肢的承重能力
- 促進肢體的抗重力運動
- 促進血管舒縮訓練

Key Steps 重點步驟

1. The transferred patient lay flat on a standing bed
2. Relax and lie on your back
3. Try to use towels and pillows to support the patient's deformed limbs
4. First, firmly stabilize the pelvis
 - 6-inch wide pelvic belt
 - The upper third and lower two-thirds of the pelvis are covered by ASIS
5. Then fix the calves
 - 4-inch wide calf strap
6. Finally, fix the position near the knee joint (avoid pressing on the kneecap).
 - 4-inch wide knee strap

1. 轉移病人平趟在企床上
2. 放鬆及仰臥
3. 盡量利用毛巾及枕頭支持病人變形的肢體
4. 先穩固盆骨
 - 6吋闊盆骨帶
 - 上1/3 及下2/3蓋盆骨的ASIS上
5. 再固定小腿
 - 4吋闊小腿帶
6. 最後固定近膝關節下的位置(避免壓在膝蓋骨上)
 - 4吋闊膝帶



<https://www.youtube.com/watch?v=rheJbel8vSk>



<https://www.youtube.com/watch?v=xnkhqXzFmN4>

Hospital bed 醫療床

Safe & protect both caregivers & disabled individuals. Prevent fall for disabled ones.

Easier for Caregivers/ carers on home care for their disabled members esp transfer.

Prolonged home care instead of institutional care.

More mobile than regular bed.

安全並保護護理人員和病人。防止病人跌倒。

護理人員更容易為病人進行家庭護理，尤其是轉移。

長期的家居護理而不是機構護理。

比普通床更靈活。



Ripple Bed 氣墊床

Aim

- **minimize risk of pressure sore**
- **reduce pressure** between patient's skin and support medium
- **distribute weight** across a large contact area

Target : individuals who are dependent on bed/ wheelchair for a long period.

目的

- 最大限度地**降低壓瘡**的風險
- **減少**患者皮膚和支撐物之間的**壓力**
- 將**重量分佈**在較大的接觸區域

對象：長期依賴床/輪椅的人士。



Change positions REGULARLY
定期更換姿勢

Rehab equipment for ADL

ADL 的康復設備

Commode 便椅

- act as **mobile toilet**
- for those with disability/ illness situations, who need assistance in toileting
- **easier transport** for caregiver on **bathing** task
- 充當**移動廁所**
- 對於殘疾/疾病情況，需要協助如廁的人
- 護理人員更**容易移動**進行洗澡



Support on Body parts 身體部位支撐

Shoulder Sling/Arm Sling

- **Protect arm** after injury to shoulder/ elbow.
- **Facilitate healing** since keep position + avoid movements
- After shoulder operation eg rotator cuff
- After stroke to **prevent subluxation**



肩背帶/手臂吊帶

- 肩膀/肘部受傷後**保護手臂**。
- **促進癒合**，因為保持姿勢+避免移動
- 肩部手術後，例如肩袖肌肉
- 中風後**預防半脫位**

Carpal Tunnel Wrist Brace 腕管腕帶

Aim

- stabilize wrist in "Neutral" position so to minimize pressure on the median nerve
- "rest" from movements
- reduce pain & numbness at affected palm & fingers
- could wear during sleep



目的

- 將手腕穩定在“中立”位置，以盡量減少對正中神經的壓力
- 從運動中“休息”
- 減輕受影響手掌和手指的疼痛和麻木
- 睡覺時可以佩戴



Back Brace 背部支架(腰封)

- Add "Stability" with better awareness on the posture.
- Limit movement.
- Minimize risk of injury eg during patient transfer/ other manual handling tasks.

**** NOT for continuous wearing**

- 添加"穩定性"，更好地了解姿勢。
- 限制運動。
- 最大限度地降低受傷風險，例如在患者轉移/其他手動搬運任務期間。

**** 不適合連續佩戴**



Reference

https://www.physio-pedia.com/Assistive_Technology_for_Positioning

https://www.physio-pedia.com/Moving_and_Handling